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# RECOVERING HIERARCHICAL STRUCTURES IN PSYCHOLOGICAL MEASUREMENT

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## Abstract

Psychological constructs are often organized hierarchically, making accurate identification of their narrowest dimensions essential to recover broader levels of organization. Existing network methods for this task are limited by either the resolution limit of modularity or the variability introduced by terminating community detection prematurely. We introduce Pairwise Resolution Iteration via Subgraph Modularity (PRISM), a community detection procedure designed to identify dimensions below the resolution limit without requiring a tunable parameter. We evaluated PRISM using 320,000 samples generated from second order factor models across 3,200 conditions that varied the numbers of first and second order factors, factor loadings, indicators per factor, sample size, skew, overlapping dimensions, population error, and correlations among second order factors. PRISM was compared with lower order Louvain and exploratory factor analysis (EFA) paired with several traditional dimensionality methods, many of which set the depth of the backward estimation procedure. Recovery was evaluated using dimensionality accuracy, bias, and structural accuracy. PRISM generally provided the most accurate and least biased recovery at both levels and maintained strong performance at smaller sample sizes and extreme skew. Overlapping dimensions reduced performance across methods, and both network methods performed poorly when overlapping dimensions were combined with few indicators, weak first order loadings, and strong second order loadings. EFA paired with parallel analysis based on principal component eigenvalues was the strongest traditional alternative, particularly in exceptionally large samples. These findings support PRISM as a robust general procedure to recover each level of a hierarchical structure.

**Keywords** hierarchical structure, dimensionality estimation, network psychometrics, resolution limit

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# 1 Introduction

Hierarchical frameworks are prevalent across psychology with notable examples including the Big Five in personality (Markon et al., 2005; Samo et al., 2026) and the Hierarchical Taxonomy of Psychopathology (HiTOP) in psychopathology (Forbes et al., 2024; Kotov et al., 2017). These frameworks span multiple research domains, often encompassing entire areas of study (e.g., psychopathological disorders). An accurate representation of these hierarchical frameworks has major consequences for theory and intervention research. Despite these consequences, simulation studies evaluating the accuracy of exploratory methods that estimate hierarchical structures has been sparse.

To our knowledge, only one study has evaluated the recovery of hierarchical structures using exploratory estimation methods (Jiménez et al., 2025), finding that the network psychometrics approach, hierarchical exploratory graph analysis (hierEGA), performed as well as or better than traditional factor analytic approaches. Beyond the dearth of investigations, there are several challenges that limit our understanding of how each level of a hierarchy can be accurately recovered. Some challenges exist in the data themselves. Psychopathology, for example, has symptom response patterns that tend to be highly skewed (Hellton et al., 2021) and some symptoms indicate multiple disorders (e.g., overlapping symptoms; Cramer et al., 2010). Other challenges exist in the methods. Although hierEGA was found to be the most accurate exploratory hierarchical estimation method in a previous simulation study (Jiménez et al., 2025), there are known limitations of the algorithms currently used in it that could prevent it from detecting dimensions below a certain size (Fortunato & Barthélemy, 2007).

The present paper aims to address each of these issues. First, we developed a new algorithm that we call Pairwise Resolution Iteration via Subgraph Modularity (PRISM) that is designed to overcome the limitations of the clustering algorithms used in hierEGA. Second, we performed a simulation study to evaluate PRISM against two contemporary standards to estimate hierarchical structures, hierEGA with lower order Louvain and parallel analysis with exploratory factor analysis (EFA; Horn, 1965; Jiménez et al., 2025). In addition, we evaluated several traditional dimensionality estimation methods that have been proposed to set the hierarchical depth of the increasingly popular bass-ackward method (Forbes, 2023; Girard, 2026; Goldberg, 2006). Finally, our simulation generated ordinal data with varying skew and first order dimensions with overlapping indicators (variables loading on to multiple dimensions), mirroring more realistic patterns observed in psychological structures.

## 1.1 Contemporary Methods to Estimate Hierarchical Structures

Factor analytic and network psychometric approaches have been the primary methods used to estimate hierarchical structures in the last decade. Although several approaches for factor analysis exist (Markon, 2019), bass-ackward (Goldberg, 2006) has become a common exploratory approach to estimate hierarchical structure of personality and psychopathology (Forbes et al., 2024, 2025, 2026; Goldberg, 2006).

The bass-ackward approach begins by setting the number of dimensions at the lowest level, which are usually suggested by several dimensionality estimation methods (Girard, 2026). The procedure then uses principal component (PCA) or exploratory factor analysis (EFA) to extract the first component or factor as the top level of the hierarchy. An additional component or factor is extracted in each subsequent level down to the number of dimensions suggested by a traditional dimensionality estimation method. Goldberg (2006) also suggested an approach that does not rely on a traditional dimensionality method where dimensions are extracted until a component or factor does not have a variable with its highest loading on it (Tong et al., 2025). Others have offered a refinement step where correlations between components or factors greater than 0.90 are dimensions extending into the next level, and correlations between 0.30 and 0.90 are new dimensions (Loehlin & Goldberg, 2014).

More recently, the extended bass-ackward approach was introduced to further develop bass-ackward including three refinements (Forbes, 2023). The first refinement was to remove dimensions of the hierarchy with correlations greater than 0.90 and consider them redundant rather than allowing them to perpetuate through the hierarchy. Instead, the dimension is retained at the lowest level that it appears. An additional heuristic is to use Tucker’s congruence coefficient (Lorenzo-Seva & Ten Berge, 2006; Tucker, 1951) greater than 0.95 between the loadings to support the identification of redundant dimensions. The second refinement was to remove dimensions that are likely statistical artifacts using theoretical knowledge. The third and final refinement was to examine the associations between dimensions across all levels rather than within a given level to better understand the relationships between broad and narrow dimensions. Importantly, these refinements do not increase the depth that bass-ackward estimates down to. Therefore, whether using the

traditional or extended bass-ackward approach, the maximum number of lower order dimensions used in bass-ackward depends *entirely* on the initial dimensionality estimate from traditional procedures.

In contrast to the top-down approach of bass-ackward, hierEGA is a bottom-up network psychometrics procedure based on exploratory graph analysis (Golino & Epskamp, 2017; Golino et al., 2020). The approach starts by first estimating a network using the graphical least absolute shrinkage and selection operator (Friedman et al., 2008) with extended Bayesian information criterion (Chen & Chen, 2008) for model selection (EBICglasso; Epskamp and Fried, 2018; Foygel and Drton, 2010). The next step applies a community detection algorithm to the network to determine the number and content of the communities (Fortunato, 2010).

Although there are many algorithms that could be used, the Louvain algorithm (Blondel et al., 2008) is favored because its procedure naturally builds a hierarchical structure. The algorithm starts by treating every variable as its own community and merges variables one-by-one based on the largest gain in modularity (Newman, 2006), a quality function that quantifies the extent to which nodes in the network are more connected within their own community than other communities. Once the algorithm completes a pass over each variable once, nodes in the same communities are merged and their connections between communities are updated. This procedure proceeds iteratively until modularity can no longer be improved or one community remains. Each pass can be treated as a level in a hierarchy with the first pass being useful for the lowest level of dimensions and has been referred to as lower order Louvain (Jiménez et al., 2025). With the dimensions from the lower order Louvain, network or factor scores are computed and are used to estimate EGA for the next level (Jiménez et al., 2025; Samo et al., 2026).

A prior simulation study by Jiménez et al. (2025) simulated bifactor structures with a varying number of general and specific number of factors and compared hierEGA with lower order Louvain against several traditional dimensionality estimation methods with EFA. They found that hierEGA recovered lower and higher order dimensions as well as or better than all methods included in their study. This study, however, generated data from normally distributed continuous variables and had a limited number and magnitude of overlapping dimensions. The accuracy of these methods to recover hierarchical structures when data are ordinal with varying skew and substantial overlapping dimensions remains an open question.

## 1.2 Resolution Limit of Modularity

The function of lower order Louvain is practical as much as it is convenient. A well-known constraint of modularity-based algorithms is that they suffer from a so-called resolution limit where they cannot detect communities below a certain size relative to the total network size (Fortunato & Barthélemy, 2007). The practical utility of the lower order Louvain algorithm lies in stopping before any community aggregation occurs, retaining smaller communities. To gain a mechanistic understanding of the resolution limit, modularity must first be defined,

$$Q = \frac{1}{2m} \sum_{ij} \left( |w_{ij}| - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j),$$

where  $i$  and  $j$  represent variables  $i$  and  $j$ ,  $\mathbf{W}$  is the network,  $\mathbf{k}$  is a vector of each node's strength or the sum of its connections ( $\sum_j |W_i|$ ),  $m$  is the total sum of the network ( $\sum_{ij} |w_{ij}|$ ),  $\mathbf{c}$  is a vector of community memberships, and  $\delta(c_i, c_j)$  is zero unless  $c_i = c_j$  where it is one. The term,  $\frac{k_i k_j}{2m}$ , represents the expected edge weights under the configuration model (Newman, 2006; Newman & Girvan, 2004), a null model that preserves each node's strength while randomizing connections. This term scales with the total edge weight of the network ( $2m$ ), which is the cause of the limit.

Fortunato and Barthélemy (2007) showed that two communities,  $A$  and  $B$ , will be merged under modularity maximization whenever

$$W_{AB} > \frac{k_A k_B}{2m},$$

where  $W_{AB} = \sum_{i \in A} \sum_{j \in B} |w_{ij}|$  is the total edge weight between communities  $A$  and  $B$  and  $k_A = \sum_{i \in A} k_i$  is the total node strength within community  $A$  ( $k_B$  defined analogously). Because the right-hand side of this condition is scaled by  $2m$ , the amount of between-community connectivity two substructures can tolerate before being merged shrinks as the network grows.

A large community identified through modularity maximization therefore cannot be assumed to be a single coherent community. It may conceal two substructures whose between-community weight,  $W_{AB}$ , is small in absolute terms yet still exceeds this shrinking threshold. The consequence is that smaller communities, even when internally well-defined, do not contribute meaningfully to global modularity relative to this threshold, leading algorithms to favor larger communities. Because  $2m$  tends to scale with the number of variables under roughly constant edge density, the resolution limit tends to appear more consistently in larger networks.

The early termination of the Louvain algorithm is therefore a convenient short circuit of its modularity maximization in the aggregation steps. This benefit comes at the cost of greater variability. The standard Louvain algorithm is already stochastic such that its output depends on the initial ordering of the nodes (Lancichinetti & Fortunato, 2012). By terminating after one pass, the lower order Louvain removes the subsequent aggregation and merging steps that could otherwise allow different initial orderings to converge on more consistent partitions.

This variability is compounded when a variable sits close to the boundary between two comparably good community assignments—a condition that finite-sample fluctuation can produce, and that overlapping dimensions may exacerbate further (Good et al., 2010). Consensus clustering (Lancichinetti & Fortunato, 2012), where a stochastic algorithm is repeatedly applied to the same network, can partially offset variability arising from node ordering, but when several partitions are similarly plausible, there may not be one dominant consensus.

Existing attempts to address the resolution limit have largely focused on rescaling the null model itself. A common approach introduces a resolution parameter,  $\nu$ , that is multiplied by the null model term to give  $\nu \frac{k_i k_j}{2m}$  (Reichardt & Bornholdt, 2006). Larger values of  $\nu$  raise the cost of merging, favoring smaller communities. This solution introduces researcher degrees of freedom because  $\nu$  must be selected. One strategy has been to select  $\nu$  by scanning a range of values and identifying partitions that remain consistent across many values of  $\nu$  (Arenas et al., 2008).

Granell et al. (2012) instead proposed a hierarchical multiresolution approach that addresses the scale dependence of the resolution limit by recursively applying multiresolution community detection to the subgraphs identified at coarser resolutions. The procedure first identifies an optimal partition of the entire network at a given resolution and then treats each resulting community as an independent subgraph, repeating the multiresolution analysis within each subgraph until no further subdivision is detected. By reducing the size of the network considered at each subsequent level, structures that may be undetectable because of the resolution limit at the level of the full network can become detectable within their corresponding subgraphs. Thus, rather than aggregating partitions obtained from an exhaustive global sweep of  $\nu$ , the approach localizes the resolution parameter to progressively smaller portions of the network.

### 1.3 Pairwise Resolution Iteration via Subgraph Modularity

Our proposed method builds on the approach by Granell et al. (2012). We begin by applying the standard Louvain algorithm to the entire network to establish an initial partition, without making any prior assumptions about potential substructures. For each community, a subgraph is first created from the community by itself, and then from its union with every other community, one at a time. Optimizing modularity on an isolated community, as suggested by Granell et al. (2012), is a well-known but overly liberal test for substructure, tending to identify multiple communities even when there is only one ground truth community (Christensen, 2024; Golino et al., 2020; Newman & Girvan, 2004). PRISM purposefully exploits this tendency as a minimum condition to detect substructure where a gain in the isolated subgraph is necessary, but not sufficient, evidence of substructure.

For each subgraph, we apply the standard Louvain algorithm and compare the resulting modularity to the modularity of the community’s current partition within that subgraph. Once every subgraph has been evaluated, we identify any community that achieves a modularity gain across *every* subgraph in which it participates—its own isolated subgraph and its union with each other community. We then update that community to reflect the partition of its own nodes that were most frequently observed across those subgraphs. In the event of a tie, we select the partition that yields the largest modularity gain, evaluated on the community’s own nodes so that the comparison is not confounded by differences in composition across the subgraphs the tied partitions came from. This entire procedure iterates until no further updates to any partition occur.

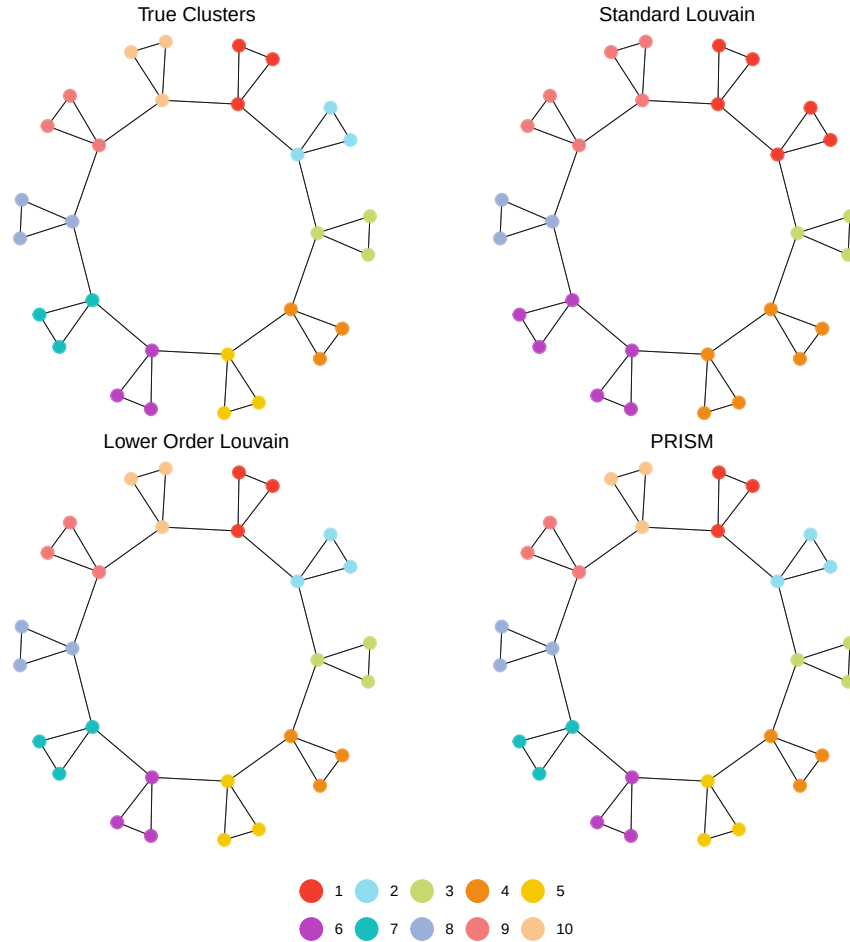
When this process converges without updating the initial partition, PRISM returns the standard Louvain partition unchanged, indicating no evidence of substructure. When no substructure exists, PRISM should

match the recovery of standard Louvain, which has been shown to be one of the most accurate community detection algorithms for recovering non-hierarchical structures in psychology (Christensen et al., 2024; Gates et al., 2016). This feature makes PRISM a general purpose algorithm rather than one limited to hierarchical structures.

The key insight to understand the effectiveness of this approach comes from recognizing that an induced subgraph reduces the effective  $2m$  in the null model relative to the full network. This rescaling allows PRISM to identify communities at a finer scale than global modularity maximization would otherwise permit, without introducing another parameter (e.g.,  $\nu$ ) or terminating before the objective criterion is met fully (e.g., lower order Louvain).

PRISM possesses two advantages over lower order Louvain by guarding against oversplitting and reducing variability associated with ordering. First, requiring a community to show unanimous modularity gains across every subgraph it participates in imposes a conservative criterion that reduces the likelihood of oversplitting. Second, because PRISM applies the standard Louvain algorithm within each subgraph rather than terminating after one pass, PRISM retains the subsequent aggregation and merging steps that lower order Louvain forgoes, allowing different initial node orderings to converge on more consistent partitions.

**Figure 1**  
*Fortunato & Barthélemy (2007) Ring Community Network*



*Note.* The true (top left), standard Louvain (top right), lower order Louvain (bottom left), and PRISM (Pairwise Resolution Iteration via Subgraph Modularity; bottom right) community memberships are depicted with colors representing same community memberships.

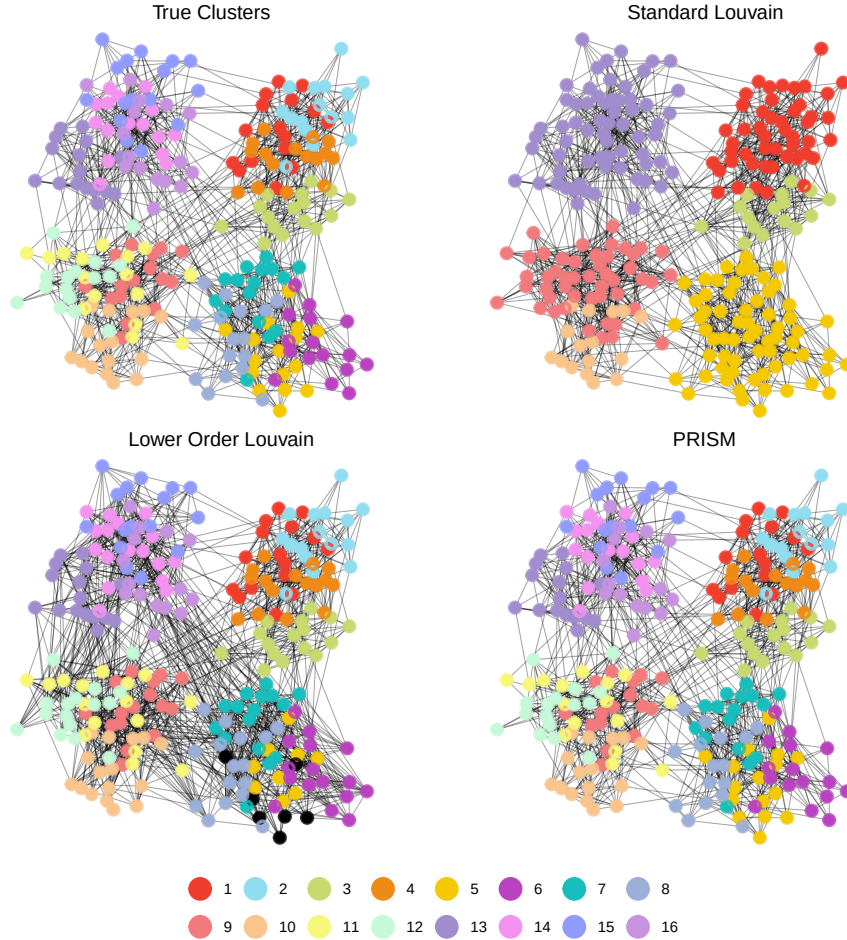
To demonstrate these issues and the effectiveness of PRISM, it was applied along with the standard Louvain and lower order Louvain to two synthetic networks commonly used to demonstrate the resolution limit. The

first network comes from the seminal resolution limit paper by Fortunato and Barthélemy (2007) who showed that a ring of communities, each connected by a single edge, cannot be adequately recovered by modularity optimizing algorithms (Figure 1).

The ring network has 10 true communities, which the lower order Louvain and PRISM algorithm correctly identify. The standard Louvain, however, identifies 6 communities, merging several of the smaller communities. This example demonstrates that the standard Louvain is insufficient to detect all substructures below the resolution limit while the lower order Louvain and PRISM can adequately recover these structures.

**Figure 2**

*Arenas et al. (2008) Hierarchical Community Network*



*Note.* The true (top left), standard Louvain (top right), lower order Louvain (bottom left), and PRISM (Pairwise Resolution Iteration via Subgraph Modularity; bottom right) community memberships are depicted with colors representing same community memberships. The black nodes represent the 17<sup>th</sup> community identified by the lower order Louvain.

The second example comes from Arenas et al. (2008) who constructed a hierarchical network structure consisting of 4 groups of second level communities that were each comprised of 4 groups of first level communities with 16 nodes each (256 nodes in total; Figure 2). The structure for this network therefore has 16 first level communities and 4 second level communities. Reasonably, an accurate algorithm could identify either the first or second level communities. An algorithm unable to resolve the resolution limit, however, would not identify the 16 first level communities. The standard Louvain algorithm most clearly demonstrates this issue, only identifying 6 communities. Of these, 2 correspond to actual second level communities. The remaining 4 arise because it fragments the other 2 second level communities, splitting each into 2 pieces: one piece that

exactly recovers an actual first level community, and one piece that merges together the rest of the first level communities belonging to that same second level community.

The lower order Louvain was much closer to the truth than the standard Louvain, identifying 17 communities. The additional community occurred from a split of a first level community. A greater issue for the lower order Louvain lurks beneath the surface. The algorithm was reapplied over 100 different seeds, recovering a wide range of solutions (frequency in parentheses): 15 (3), 16 (13), 17 (29), 18 (28), 19 (7), 20 (11), 21 (6), and 22 (3). The standard Louvain showed far less variability across seeds (frequency in parentheses): 4 (6), 5 (47), and 6 (47), but only by merging away the very substructure the lower order Louvain was intended to detect. PRISM suffered from neither limitation, identifying the true 16 community structure in all 100 seeds.

## 2 Present Research

The present study evaluates the effectiveness of PRISM to recover hierarchical structures commonly found in psychological measurement. PRISM is compared against several alternatives including hierEGA with lower order Louvain and EFA paired with traditional dimensionality estimation methods, such as parallel analysis with principal axis factoring eigenvalues (PAPaf) and parallel analysis with principal component eigenvalues (PAPca). All of the traditional dimensionality estimation methods evaluated, including Goldberg’s maximum loading criterion (Goldberg, 2006; Tong et al., 2025), are used as a depth-setting heuristic for the increasingly popular bass-ackward method (Girard, 2026). All of these methods were compared at the first order level for their recovery of the number of dimensions and structure (item assignments). PRISM, hierEGA with lower order Louvain, PAPaf, and PAPca were additionally used to estimate the second order level. This application was based on the prior precedent of reapplying each method to the lower order network or factor scores to determine the higher order structure (Jiménez et al., 2025; Samo et al., 2026).

Our simulation design differed from the work of Jiménez et al. (2025) in important ways. First, all data were generated as polytomous (5-point Likert scale) with moderate and extreme skew to assess recovery under data conditions found across psychological applications (Hellton et al., 2021; Huth et al., 2025). Second, items had substantial cross-loadings between 0.40–0.50 to mirror patterns observed in overlapping dimensions such as those in psychopathology (Cramer et al., 2010). Third, we tested the recovery of the structure in addition to the accuracy of dimensionality estimation. Similar to Jiménez et al. (2025), we simulated population error or perturbations added to the population correlation matrix to emulate model misspecification to evaluate structure recovery under more realistic conditions (Cudeck & Browne, 1992; Kracht & Waller, 2025).

Importantly, we included a single second order factor condition to test whether PRISM can recover structures that could be equally represented as a correlated factor model—structures that are consistent with non-hierarchical simulations (e.g., Christensen et al., 2024). Recovery was evaluated separately at the first and second order levels using three complementary metrics: exact match accuracy and bias for the number of dimensions and the Omega index (Collins & Dent, 1988), a measure of structure recovery that generalizes the adjusted Rand index (Hubert & Arabie, 1985) to accommodate overlapping dimensions.

## 3 Method

### 3.1 Simulated Conditions

All simulated data were polytomous (5-point Likert scale) and generated from a second order factor model. The number of second order (1, 2, 4, and 6) factors were selected to sample from narrower to broader hierarchical structures, with the highest level (6) reflecting the number of higher order dimensions in personality and psychopathology (Forbes & Wright, 2023; Samo et al., 2026). The number of first order factors per second order factor was 3 and 5, making the total first order factors range from 3 to 30. Primary factor loadings for the first and second order factors were both manipulated to be 0.50 (moderate) and 0.70 (large). Correlations for the second order factors were set to small (0.10), moderate (0.30), and large (0.50). The correlations between the first order factors were implied by the second order loadings and correlations. Sample sizes were generated using 500 (small), 1,000 (moderate), 2,500 (large), 5,000 (very large), and 10,000 (exceptionally large). Manipulated parameters of particular interest were skew, overlapping dimensions, and population error. Skew was manipulated to be either moderate or extreme whereas overlapping dimensions and population error were specified as either present or absent.

For the unidimensional second order factor, the design was fully crossed with  $2 \times 2 \times 2 \times 5 \times 2 \times 2 \times 2$  (first order factors  $\times$  first order loadings  $\times$  second order loadings  $\times$  sample size  $\times$  skew  $\times$  overlapping dimensions  $\times$  population error), resulting in 320 conditions. The multidimensional second order factors were similarly crossed with the addition of correlations between second order factors with  $320 \times 3 \times 3$  (all previous conditions  $\times$  second order factors  $\times$  second order factor correlations), resulting in 2,880 conditions. In total, 3,200 conditions were generated with 100 replicates each, totaling 320,000 replicate samples.

### 3.2 Data Generation

We generated data from multivariate normal second order factor models by first computing the reproduced population correlation matrix:

$$\mathbf{R}_R = \mathbf{\Lambda}_F \mathbf{\Phi}_F \mathbf{\Lambda}'_F + \mathbf{\Psi}_F$$

and

$$\mathbf{\Phi}_F = \mathbf{\Lambda}_S \mathbf{\Phi}_S \mathbf{\Lambda}'_S + \mathbf{\Psi}_S$$

where  $\mathbf{R}_R$  is the reproduced population correlation matrix, subscripts F and S represent first order and second order levels (respectively),  $\mathbf{\Lambda}$  represents a loading matrix for a given level,  $\mathbf{\Phi}$  represents a factor correlation matrix for a given level, and  $\mathbf{\Psi}$  represents the disturbances for a given level. For our procedure,  $\mathbf{\Psi}_S$  was set as an identity matrix (no residual variance) and  $\mathbf{\Phi}_F$  was implied by  $\mathbf{\Lambda}_S \mathbf{\Phi}_S \mathbf{\Lambda}'_S$ . The population correlation matrix,  $\mathbf{R}_P$ , was obtained by putting unities on the diagonal of  $\mathbf{R}_R$  and performing a Cholesky decomposition such that

$$\mathbf{R}_P = \mathbf{U}'\mathbf{U}.$$

#### 3.2.1 Overlapping Dimensions

Overlapping dimensions were created by adding substantial cross-loadings to first order factors that loaded onto the same second order factor in a cyclical pattern (e.g.,  $1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 1$ ). To avoid creating Heywood cases, the cross-loading and primary loading (replacing the original value) were generated from a uniform distribution between 0.40 and 0.50. The proportion of cross-loading items per factor was set to 0.20 (1 and 2 cross-loaded variables per factor for 5 and 10 variables per factor, respectively).

#### 3.2.2 Population Error

Population error was added to  $\mathbf{R}_P$  following the Tucker, Koopman, and Linn method (Tucker et al., 1969), which adds numerous minor factors with small effects to introduce random sources of error into the population correlation matrix. Specifically,

$$\mathbf{R}_P = \mathbf{\Lambda}_F \mathbf{\Phi}_F \mathbf{\Lambda}'_F + \mathbf{W}\mathbf{W}' + \mathbf{\Psi}_F,$$

where  $\mathbf{W}$  is a  $p \times q$  matrix of minor factor loadings, with  $q$  representing the number of minor factors (Kracht & Waller, 2025), and  $\mathbf{\Psi}_F$  is a diagonal matrix of unique variances scaled such that  $\mathbf{R}_P$  has a unit diagonal. Our study used the default number of minor factors ( $q = 50$ ) in the `noisemaker` function in the R package {fungible} (version 2.4.8; Waller, 2026).

Because population error introduces small disturbances, cross-loadings at the first and second order levels are implicitly generated. The cross-loadings in the first ( $\mathbf{\Lambda}_F$ ) and second ( $\mathbf{\Lambda}_S$ ) order loading matrices were therefore set to zero, implying simple structure prior to the addition of population error, except in conditions with overlapping dimensions.

To calibrate population error, the `noisemaker` function offers the comparative fit index (CFI; Bentler, 1990) and root mean square error of approximation (RMSEA; Steiger, 1990). Nguyen and Waller (2023) showed that when population error was calibrated using CFI, models classified as having good or moderate fit also had acceptable RMSEA values. In contrast, when population error was calibrated using RMSEA, many models classified as well-fitting by RMSEA had CFI values below 0.90. In this regard, RMSEA has been shown to be overly sensitive to model size, indicating improved model fit as the number of variables increase regardless

of misspecification (Kenny & McCoach, 2003; Shi et al., 2019). Conversely, CFI has been demonstrated to be more robust to incidental parameters (Garrido et al., 2016). We therefore calibrated population error using CFI.

A preliminary simulation was conducted across the full set of structural conditions (excluding sample size) to determine the amount of population error that could be introduced while maintaining the recovery of the specified factor structure in the population using exploratory factor analysis (EFA) with the correct number of dimensions. Population correlation matrices were generated with increasing levels of error, which was controlled by a target level of (mis)fit indexed by CFI, ranging from 0.99 to 0.95 in decrements of 0.01. For each level of error, we evaluated whether the simulated first and second order structures could be recovered, including conditions with overlapping dimensions. Based on these results, a CFI of 0.97 was selected as the maximum level of misfit that had consistently recoverable structures across the simulated conditions.

### 3.2.3 Convergence Criteria

Several criteria were used to ensure the quality of the correlation matrices and subsequent data generation. The first order factor correlation matrix ( $\Phi_F$ ) and population correlation matrix ( $R_P$ ) were required to be positive definite, with no communalities exceeding 0.90. The population model was also required to meet the target level of misfit ( $|CFI - 0.97| \leq 0.01$ ), have no residual correlations greater than 0.10 (Shi et al., 2018), and permit recovery of the first and second order factor structures in the population. If any criterion was not satisfied,  $R_R$  was regenerated and the procedure was repeated until all criteria were met.

### 3.2.4 Skew Generation

Samples of continuous data were generated using,  $X = ZU$ , where  $Z$  is a matrix of random multivariate normal data with rows equal to the sample size and columns equal to the total number of variables.

To convert the continuous data to polytomous variables, category probabilities were selected to produce the desired level of positive skewness. Following the general approach of Garrido et al. (2011), the first category was assigned to an increasingly large probability, with progressively smaller probabilities assigned to the remaining categories. The rate at which these probabilities decreased was controlled by a reduction factor. For each set of category probabilities, the resulting skewness was calculated directly from the expected number of responses in each category. The algorithm then searched for the probability assigned to the first category that produced the target skewness within a tolerance of  $1 \times 10^{-5}$ .

The search began with a reduction factor of 0.75 and  $N = 1,000,000$ . If the target skewness could not be reproduced closely enough, larger sample sizes were used to provide finer differences among possible category probabilities. When increasing the sample size was insufficient, other values for the reduction factor were searched. Finally, the cumulative category probabilities were converted to  $z$ -score thresholds using the inverse standard normal cumulative distribution function, and these thresholds were used to divide normally distributed variables into the desired response categories. This procedure was used to produce pregenerated skew tables, with increments of 0.05 skewness, in the {latentFactor} (version 0.0.9; Christensen et al., 2026) package in R.

Moderate skew was generated from a uniform distribution ranging from 1 to 2. To generate more extreme skew values that reflected those observed in areas such as psychopathology, we generated skewness based on real data from two large psychopathology inventories. The first inventory was the Symptom Checklist-90-Revised (Derogatis, 1994) completed by a combined psychiatric and community sample ( $N = 2,069$ ; Arrindell et al., 2026). The second inventory was the HiTOP Self-Report (Hierarchical Taxonomy of Psychopathology Society, 2024) completed by a sample of cancer survivors ( $N = 728$ ; Haywood et al., 2026). The item-level skew values from these datasets were used to identify a best fitting distribution from which to generate skew values.

Several probability distributions were fit by maximum likelihood to the absolute item-level skew values from each inventory. The gamma distribution provided the best fit in both samples, with estimated shape parameters of 2.14 and 1.27 and rate parameters of 1.25 and 0.75, respectively. With the goal of representing broad psychopathology inventories that are likely to span these distributions, we generated skew values from a gamma distribution with the average shape and rate parameters (1.70 and 1, respectively). Because no skew values from either sample were greater than 10, if any single skew value in the population was larger than 10, we regenerated skew with the same distribution and parameters until all skew values in the population were less than 10. The extreme skew in the generated data had a mean of 1.70, median of 1.38, standard deviation of 1.30, and range between -0.052 and 12.55.

### 3.3 Estimation Methods

All estimation methods used polychoric correlations to estimate the correlation matrices from the item-level data. For extreme skew conditions, there is a possibility that variables could collapse to two categories with empty cells. We implemented empty cell corrections following contemporary recommendations where 0.50 is added to empty cells of  $2 \times 2$  tables only and no correction for larger tables (Savalei, 2011; Yang & Weng, 2024).

#### 3.3.1 First and Second Order

hierEGA with PRISM and lower order Louvain as well as the two parallel analysis methods were used to estimate both the first and second order structures.

hierEGA followed two paths: one using the newly developed approach, PRISM, and the other using the standard approach using lower order Louvain (Jiménez et al., 2025). Following the hierEGA procedure, a unidimensional check is performed prior to estimating a network at each level (Golino et al., 2020). If the estimated correlation matrix at the given level is determined to be unidimensional by the standard Louvain algorithm, then the result for that level is one dimension (Christensen, 2024); otherwise, a network is estimated.

Both approaches estimate a network using the EBICglasso method from the correlations. The  $\lambda$  parameter was set by conducting a grid search across 100 candidate values and selecting the value that minimized EBIC with  $\gamma$  set to 0.50. If any nodes were disconnected, then  $\gamma$  was decreased by 0.25 and the network was re-estimated. The same procedure continued until  $\gamma = 0$  where the resulting network was retained (Golino et al., 2020).

Following standard procedure (Epskamp & Fried, 2018), the search bounds for  $\lambda$  were defined as

$$\lambda_{\max} = \max_{i \neq j} |\hat{r}_{ij}|$$

and

$$\lambda_{\min} = \lambda_{\max} \cdot 0.1,$$

where  $\hat{r}_{ij}$  denotes the off-diagonal elements of the estimated correlation matrix  $\hat{\mathbf{R}}$ . The full sequence of  $\lambda$  values was obtained by the exponentiation of 100 equally spaced values between  $\log \lambda_{\min}$  and  $\log \lambda_{\max}$ , yielding a geometrically spaced grid from  $\log \lambda_{\min}$  to  $\log \lambda_{\max}$ .

For each estimated network, the PRISM and lower order Louvain were both applied with consensus clustering (Lancichinetti & Fortunato, 2012). For PRISM, the standard Louvain algorithm was applied 1,000 times and retained the most common solution as the result at the initial clustering stage as well as each pairwise subgraph. For lower order Louvain, the algorithm was applied 1,000 times and the most common solution was retained as the result.

The resulting communities for both were then used with the same estimated network,  $\hat{\mathbf{W}}$ , to estimate network loadings (Christensen et al., 2025):

$$\xi_{ic} = \frac{p_c}{p_c - \mathbb{1}[\mathbf{c} = c_i]} \sum_{j \in c} v_j \hat{w}_{ij}$$

and

$$\eta_{ic} = \frac{\xi_{ic}}{(\sum_{i \in c} |\xi_{ic}|)^{1/\log(\zeta p_c)}}$$

where  $p_c$  is the number of nodes in community  $c$ , and  $\mathbb{1}[\mathbf{c} = c_i]$  is an indicator equal to 1 when  $c$  is node  $i$ 's own community (correcting for the zero self-connection in  $\hat{w}_{ii}$ ) and 0 otherwise.  $\hat{w}_{ij}$  is the estimated (regularized) partial correlation between nodes  $i$  and  $j$ , and  $v_j \in \{-1, +1\}$  is the sign assigned to node  $j$  so that node connections within each community are oriented in a predominantly positive direction.  $\zeta$

is a scaling constant, set to 2 by default, that governs the magnitude of the standardized loadings,  $\eta_{ic}$  (Christensen et al., 2025).

The network loadings for both approaches were then used to compute composite correlations for the next level. These composite correlations were estimated using

$$\eta_{ic}^* = \eta_{ic} \cdot \mathbb{1}[\mathbf{c} = c_i],$$

$$\hat{\Sigma}_F = \boldsymbol{\eta}^* \hat{\mathbf{R}} \boldsymbol{\eta}^{*'}.$$

$$\hat{\mathbf{R}}_F = \mathbf{D}^{-1/2} \hat{\Sigma}_F \mathbf{D}^{-1/2},$$

and

$$\mathbf{D} = \text{diag}(\hat{\Sigma}_F),$$

where  $\boldsymbol{\eta}^*$  is the simple structure network loadings with the only non-zero loadings for each variable's assigned community,  $\hat{\Sigma}$  is the estimate covariance matrix, and  $\hat{\mathbf{R}}_F$  are the correlations between the first order dimensions.

$\hat{\mathbf{R}}_F$  was used to estimate the network for the first order dimensions for both methods. PRISM was applied to this network to estimate the second order structure. For lower order Louvain, the standard Louvain algorithm with consensus clustering (1,000 iterations) was applied to this network to estimate the second order structure (following Jiménez et al., 2025). Network loadings were computed for both to obtain the second order loadings.

Parallel analysis (Horn, 1965) followed the hierarchical implementation of Jiménez et al. (2025). The procedure starts by estimating parallel analysis from the item-level data by computing the empirical eigenvalues and comparing them against the eigenvalues obtained from  $N$  (e.g., 20) permuted datasets. The consecutive number of empirical eigenvalues greater than the mean of the permuted eigenvalues are the first order number of factors. Using this number of dimensions, an EFA was fit using the unweighted least squares (ULS) estimator (Krijnen, 1996) and geominQ rotation (Browne, 2001) with 100 iterations. For all geominQ rotations hereafter, the epsilon parameter was adjusted to be 0.0001 for two factors, 0.001 for three factors, and 0.01 for four or more factors (Muthén & Muthén, 1998-2017).

Thurstone factor scores (Thurstone, 1947) were then computed and used in parallel analysis using Pearson's correlation to estimate the second order number of factors. EFA was then fit on the first order factor scores using the maximum likelihood estimator and geominQ rotation with 100 iterations. Eigenvalues from principal component analysis and principal axis factoring were both computed with the latter putting the squared multiple correlation on the diagonal of the empirical and permuted correlation matrices.

### 3.3.2 First Order Only

The following methods were applied only at the first order level to evaluate the recovery of dimensionality methods used to estimate the depth of the bass-ackward method. For all of these methods (except for Goldberg's bass-ackward maximum loading procedure), loadings were obtained to estimate the structure using an EFA fit with the ULS estimator and geominQ rotation with 100 iterations.

Velicer's minimum average partial (MAP) criterion (Velicer, 1976) was estimated from the principal components of the polychoric correlation matrix. Beginning with one component, successive principal components were partialled from the correlation matrix and the average squared partial correlation was computed at each step. The number of dimensions was selected as the number of components corresponding to the minimum average squared partial correlation.

Very Simple Structure (VSS; Revelle and Rocklin, 1979) was evaluated at complexities one and two (VSS1 and VSS2). The principal components were varimax rotated and simplified such that either the largest absolute loading (VSS1) or the two largest absolute loadings (VSS2) for each item were retained and all remaining loadings were set to zero. The simplified loading matrix was used to reproduce the correlation matrix, and VSS fit was computed as the proportion of the total variance that is accounted for by the

simplified structure. The number of dimensions that maximized this fit was retained separately for VSS1 and VSS2. Candidate solutions ranged up to twice the population number of first order factors.

The maximum loading approach followed Goldberg’s (2006) description and Tong et al.’s (2025) implementation. The approach started by estimating a sequence of EFA models using minimum residual estimation and varimax rotation. For each solution, every item was assigned to the factor on which it had its largest absolute loading. The number of factors was increased until a solution was reached in which at least one estimated factor was not the largest loading factor for any item (Goldberg, 2006). The preceding nondegenerate solution was then retained as the dimensionality estimate. To reduce unnecessary computation while allowing the criterion to search around values supported by the other factor analytic methods, the search range was centered on the dimensionality estimates obtained from the two parallel analysis methods, MAP, VSS1, and VSS2, extending two dimensions below the smallest estimate and two dimensions above the largest estimate, subject to a minimum of one factor and a maximum equal to the number of variables. Because this procedure represents a direct implementation of a bass-ackward stopping criterion, the varimax loadings were carried forward (rather than re-estimating them with EFA using the ULS estimator and geominQ rotation with 100 iterations.).

### 3.4 Evaluation Metrics

Four metrics were computed to provide a different perspective of accuracy and bias for both the first and second order factors. The exact dimensionality accuracy was computed using,

$$\sum_{i=1}^n \frac{a_i}{n} \quad \text{where} \quad a_i = \begin{cases} 1 & \text{if } \hat{f} = f, \text{ then} \\ 0 & \text{otherwise} \end{cases},$$

where  $\hat{f}$  and  $f$  are the estimated and ground truth number of factors and  $n$  is the total number of replicates by a given number of conditions. This form of accuracy represents a direct interpretation of how often a given approach estimated the exact number of first or second order dimensions correctly.

Mean bias error (*MBE*) was computed to determine the direction of bias:

$$MBE = \sum_{i=1}^n \frac{\hat{f} - f}{n}.$$

A positive value indicates a method has a tendency to overfactor whereas a negative value indicates a method has a tendency to underfactor.

Mean absolute error (*MAE*) was computed to determine the absolute bias:

$$MAE = \sum_{i=1}^n \frac{|\hat{f} - f|}{n}.$$

Values around the same absolute value of the *MBE* suggests that the absolute bias was consistently in the direction suggested by MBE whereas greater deviations from the *MBE* suggests a more symmetric bias (under- and overfactoring depending on the conditions), while a zero indicates a lack of bias.

The Omega index ( $\omega$ ; Collins and Dent, 1988) was used to compute a method’s accuracy of determining the ground truth structure including overlapping dimensions. In the network literature, these overlapping dimensions have been referred to as overlapping communities (Cramer et al., 2010; Lange, 2021; Santiago et al., 2024). The full matrix of the estimated factor and network loadings were used for all methods to decide an item’s dimension assignment.

For conditions where overlapping dimensions were present, items were assigned to the dimensions where they had their largest and second largest absolute loadings (regardless of magnitude). To avoid noisy comparisons, items that were not generated to be on an overlapping dimension were only assigned to the dimension where they had their largest absolute loading. For conditions without overlapping dimensions,  $\omega$  reduces to the adjusted Rand index (Hubert & Arabie, 1985). In these conditions, items were assigned to the dimension where they had their largest absolute loading (regardless of magnitude).

Following Collins and Dent (1988),  $\omega$  generalizes the adjusted Rand index to solutions in which items may belong to more than one dimension. For  $n$  items,  $O = \binom{n}{2}$  denotes the total number of item pairs. For a given structure,  $X \in \{A, B\}$ , let  $d_{ik}^X$  denotes the number of factors to which both items  $i$  and  $k$  jointly belong with  $M_X = \max_{i,k} d_{ik}^X$  denoting the maximum number of factors they both belong to.  $O_j^A$  and  $O_j^B$  denotes the number of item pairs sharing exactly  $j$  clusters in  $A$  and  $B$ , respectively, and  $n_j$  denotes the number of pairs sharing exactly  $j$  clusters in *both*  $A$  and  $B$ . The observed and expected agreement, respectively, are then

$$\omega_u = \frac{1}{O} \sum_{j=0}^{\min(M_A, M_B)} n_j,$$

$$\omega_e = \frac{1}{O^2} \sum_{j=0}^{\min(M_A, M_B)} O_j^A O_j^B,$$

and the Omega index is given by

$$\omega = \frac{\omega_u - \omega_e}{1 - \omega_e}.$$

$\omega_u$  is the proportion of item pairs on which  $A$  and  $B$  agree with respect to the number of shared cluster memberships, and  $\omega_e$  is the corresponding agreement expected by chance. When no dimensions are overlapping (i.e.,  $M_A = M_B = 1$ ),  $j \in \{0, 1\}$  for every pair, so that  $\omega$  reduces exactly to the adjusted Rand index (Hubert & Arabie, 1985). Following Steinley’s (2004) recovery benchmarks for the adjusted Rand index,  $\omega$  values less than 0.65 were interpreted as poor recovery and values greater than 0.65, 0.80, and 0.90 were interpreted as moderate, good, and excellent recovery, respectively.

### 3.5 Transparency and Openness

All simulations, analyses, and visualizations were carried out using R (version 4.5.1; R Core Team, 2026). All data were generated using the `{latentFactorR}` package (version 0.0.9; Christensen et al., 2026). In-house versions of max loading, MAP, PApaf, PApca, and VSS were created following the `{psych}` package’s (version 2.6.5; Revelle, 2026) implementations for consistency with applied practices. All hierEGA with PRISM and lower order Louvain procedures were applied using the `{EGAnet}` package (version 2.5.0; Golino and Christensen, 2026). All visualizations were created using the `{ggplot2}` (version 4.0.3; Wickham, 2016) and `{ggpubr}` (version 0.6.3; Kassambara, 2026) packages. All data generation and simulation scripts are available on [GitHub](#).

## 4 Results

The overall results for the first order factors are broken down by overlapping dimensions, population error, and skew severity (Table 1). Across methods, recovery was poorest when skew was extreme and overlapping dimensions were present, whereas population error had a comparatively smaller effect for most methods. PRISM performed as well as or better than all other methods across all metrics in nearly all conditions. The only exception occurred when overlapping dimensions were present and skew was moderate: PApaf had slightly higher accuracy when population error was absent (87.9% vs. 85.0%). The differences in  $\omega$ , however, were trivial (0.973 and 0.970 vs. 0.952 and 0.950) and PRISM had comparable or better MBE (−0.673 and −0.678 vs. 0.566 and 1.002) and MAE (0.690 and 0.703 vs. 1.440 and 1.810). PRISM was one of the least affected methods for moving from moderate to extreme skew: accuracy dropped by less than 9%,  $\omega$  decreased by no more than 0.012, and MBE and MAE changed by less than 0.150, regardless of population error. PRISM was primarily affected by overlapping dimensions, which reduced accuracy by about 8–14%, shifted MBE by about 0.700 in the negative direction, and increased MAE by about 0.500–0.700.

Lower order Louvain was the second best method in terms of accuracy (85.3% to 92.3%) and structural recovery ( $\omega = 0.977$ – $0.992$ ) when overlapping dimensions were absent, but substantially impacted by overlapping dimensions, dropping between 15% and 19% in accuracy. PApca was the next best method with its performance less affected by overlapping dimensions (with accuracy dropping less than 6%) than the PRISM

and lower Louvain methods. Its accuracy also decreased with extreme skew (by about 4–11%), somewhat more than PRISM (by about 3–9%). PApaf was a strong performer in terms of accuracy and  $\omega$  when skew was moderate (accuracy = 81.4%–87.9%,  $\omega$  = 0.970–0.973), but it was exceptionally sensitive to skew (reducing accuracy by about 43–47% and increasing MBE and MAE by approximately 4–6). It was also the method for which population error most strongly reduced accuracy (by about 5–8%).

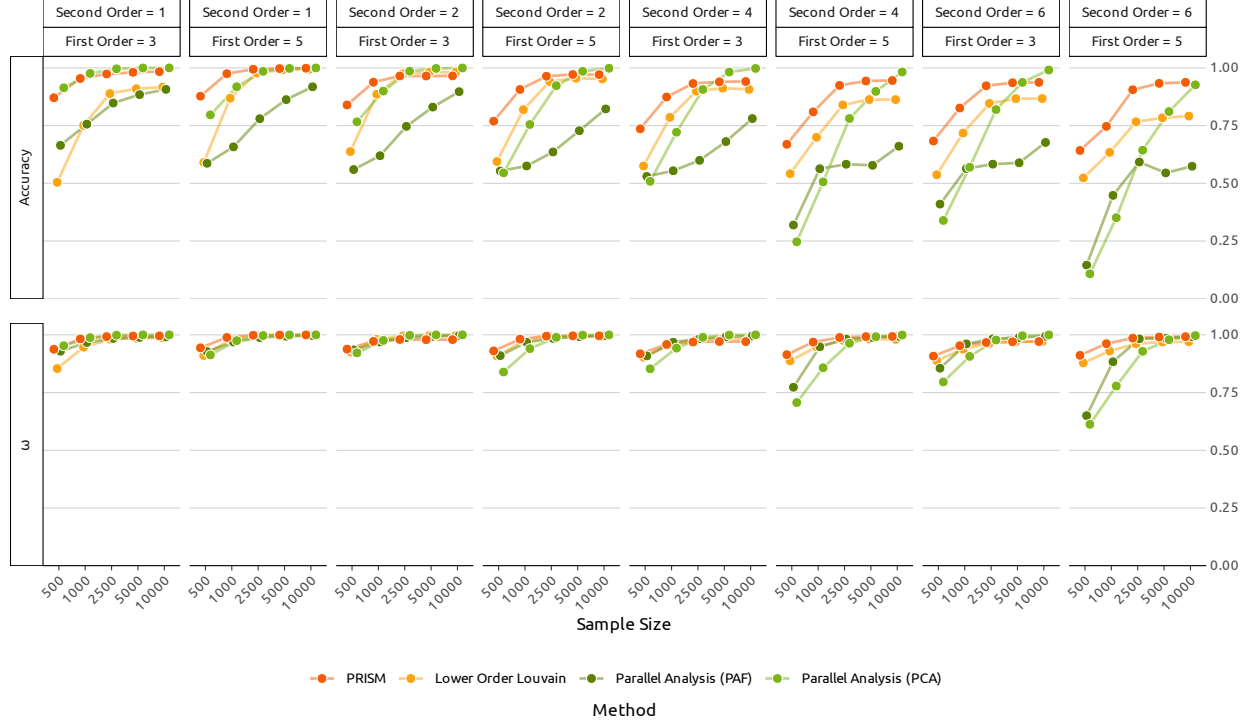
Table 1: Overall recovery of first order factors and content, by skew, error, and overlap condition

| Method                      | Overlap | Error   | Moderate Skew |               |              |              | Extreme Skew |               |              |              |
|-----------------------------|---------|---------|---------------|---------------|--------------|--------------|--------------|---------------|--------------|--------------|
|                             |         |         | Accuracy      | MBE           | MAE          | $\omega$     | Accuracy     | MBE           | MAE          | $\omega$     |
| PRISM                       | Absent  | Absent  | <b>0.988</b>  | <b>-0.001</b> | <b>0.016</b> | <b>0.996</b> | <b>0.903</b> | <b>0.142</b>  | <b>0.160</b> | <b>0.984</b> |
|                             |         | Present | <b>0.987</b>  | <b>-0.002</b> | <b>0.019</b> | <b>0.995</b> | <b>0.903</b> | <b>0.143</b>  | <b>0.162</b> | <b>0.983</b> |
|                             | Present | Absent  | 0.850         | -0.673        | <b>0.690</b> | 0.952        | <b>0.822</b> | <b>-0.568</b> | <b>0.679</b> | <b>0.944</b> |
|                             |         | Present | <b>0.846</b>  | <b>-0.678</b> | <b>0.703</b> | 0.950        | <b>0.816</b> | <b>-0.575</b> | <b>0.699</b> | <b>0.942</b> |
| Lower Order<br>Louvain      | Absent  | Absent  | 0.923         | 0.118         | 0.133        | 0.992        | 0.861        | 0.301         | 0.316        | 0.978        |
|                             |         | Present | 0.914         | 0.133         | 0.148        | 0.991        | 0.853        | 0.320         | 0.334        | 0.977        |
|                             | Present | Absent  | 0.739         | -0.851        | 0.965        | 0.942        | 0.708        | -0.707        | 0.977        | 0.931        |
|                             |         | Present | 0.726         | -0.808        | 0.974        | 0.939        | 0.695        | -0.666        | 0.998        | 0.928        |
| Bass-Ackward<br>Max Loading | Absent  | Absent  | 0.741         | -2.451        | 2.793        | 0.878        | 0.323        | -1.207        | 3.683        | 0.854        |
|                             |         | Present | 0.735         | -2.545        | 2.915        | 0.872        | 0.483        | -1.623        | 3.464        | 0.854        |
|                             | Present | Absent  | 0.683         | -3.652        | 3.823        | 0.833        | 0.330        | -2.825        | 4.395        | 0.815        |
|                             |         | Present | 0.671         | -3.760        | 3.956        | 0.825        | 0.480        | -3.196        | 4.215        | 0.812        |
| MAP                         | Absent  | Absent  | 0.668         | -3.216        | 3.216        | 0.789        | 0.620        | -3.657        | 3.689        | 0.760        |
|                             |         | Present | 0.665         | -3.243        | 3.243        | 0.787        | 0.613        | -3.700        | 3.736        | 0.758        |
|                             | Present | Absent  | 0.697         | -2.433        | 2.433        | 0.839        | 0.645        | -2.912        | 2.945        | 0.809        |
|                             |         | Present | 0.690         | -2.493        | 2.493        | 0.835        | 0.635        | -2.989        | 3.027        | 0.804        |
| Parallel<br>Analysis<br>PAF | Absent  | Absent  | 0.859         | 1.178         | 1.768        | 0.973        | 0.394        | 6.284         | 6.916        | 0.937        |
|                             |         | Present | 0.814         | 1.802         | 2.343        | 0.970        | 0.342        | 8.160         | 8.728        | 0.930        |
|                             | Present | Absent  | <b>0.879</b>  | <b>0.566</b>  | 1.440        | <b>0.973</b> | 0.450        | 4.693         | 5.672        | 0.943        |
|                             |         | Present | 0.824         | 1.002         | 1.810        | <b>0.970</b> | 0.374        | 6.915         | 7.810        | 0.934        |
| Parallel<br>Analysis<br>PCA | Absent  | Absent  | 0.842         | -0.734        | 0.739        | 0.950        | 0.756        | -0.518        | 0.825        | 0.932        |
|                             |         | Present | 0.845         | -0.681        | 0.693        | 0.951        | 0.738        | -0.425        | 0.823        | 0.933        |
|                             | Present | Absent  | 0.788         | -1.076        | 1.077        | 0.938        | 0.750        | -1.050        | 1.132        | 0.923        |
|                             |         | Present | 0.796         | -1.009        | 1.010        | 0.940        | 0.744        | -0.952        | 1.072        | 0.925        |
| VSS1                        | Absent  | Absent  | 0.439         | -4.458        | 5.799        | 0.659        | 0.364        | -3.815        | 6.446        | 0.645        |
|                             |         | Present | 0.453         | -4.511        | 5.779        | 0.659        | 0.365        | -3.906        | 6.368        | 0.647        |
|                             | Present | Absent  | 0.216         | -7.999        | 8.493        | 0.437        | 0.160        | -7.715        | 8.879        | 0.428        |
|                             |         | Present | 0.232         | -8.035        | 8.462        | 0.438        | 0.181        | -7.787        | 8.768        | 0.431        |
| VSS2                        | Absent  | Absent  | 0.424         | -2.757        | 4.521        | 0.759        | 0.350        | -1.798        | 5.498        | 0.737        |
|                             |         | Present | 0.441         | -2.793        | 4.480        | 0.761        | 0.365        | -1.957        | 5.346        | 0.740        |
|                             | Present | Absent  | 0.366         | -4.650        | 5.198        | 0.680        | 0.315        | -4.169        | 5.710        | 0.667        |
|                             |         | Present | 0.385         | -4.680        | 5.156        | 0.681        | 0.325        | -4.295        | 5.608        | 0.668        |

*Note.* Bolded Accuracy and  $\omega$  values represent the highest value for a given overlap, error, and skew condition and metric, across all eight methods. Because MBE is a signed measure of bias, bolded MBE values represent the value closest to zero (smallest absolute magnitude) for a given condition, not the numerically highest value. Bolded MAE values represent the lowest value for a given condition. MAP = minimum average partial criterion, VSS = very simple structure, PCA = principal component analysis, PAF = principal axis factoring, MBE = mean bias error, and MAE = mean absolute error.

The remaining methods (max loading, MAP, VSS1, and VSS2) performed considerably worse across all conditions. None exceeded an accuracy of 75%, even in the least challenging conditions (moderate skew, no overlapping dimensions or population error), and all showed a pronounced tendency to underfactor by more than two factors on average (MBE =  $-1.207$  to  $-8.035$ ), with VSS1 and VSS2 performing the worst (accuracy = 16.0–45.3% and 31.5–44.1%, respectively). Because these methods performed so poorly, we do not consider them in the rest of the results.

**Figure 3**  
*First Order Accuracy and Structural Recovery*



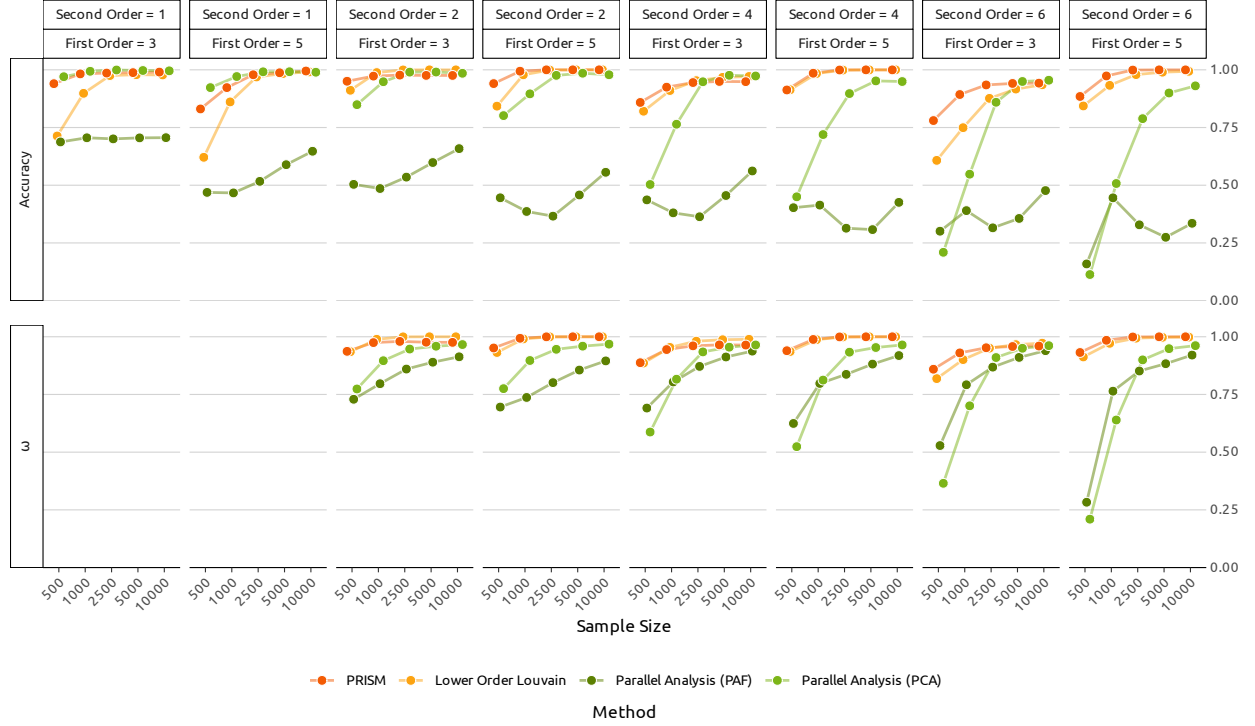
*Note.* Second Order = number of second order factors, First Order = number of first order factors per second order factor, and  $\omega$  = Omega index. A single second order factor is equivalent to a correlated factors model.

Figure 3 shows the first order factor recovery for each method broken down by the number of second order factors and the number of first order factors per second order factor. When there is a single second order factor, the structure is equivalent to a correlated factors model with no hierarchical substructure. In these conditions, accuracy increased with sample size for all methods and PRISM performed well regardless of sample size (accuracy  $\geq 80.5\%$ ). PRISM and PApca had comparable accuracy with three first order factors, but with five first order factors PRISM was considerably more accurate than PApca in smaller samples ( $N \leq 1,000$ ). Lower order Louvain performed comparably to PApca when there were five first order factors, but performed poorly in smaller samples ( $N \leq 1,000$ ) with three first order factors (accuracy  $\leq 75\%$ ). Importantly, PRISM had a near zero bias in these conditions (MBE = 0.051 and MAE = 0.063) compared to lower order Louvain (MBE = 0.249 and MAE = 0.251), suggesting that it rarely split dimensions when there was no lower order structure. PApaf had the lowest accuracy, improving with sample size but never exceeding 90% even at exceptionally large sample sizes ( $N = 10,000$ ). Structural recovery was good for all methods with a single second order factor ( $\omega \geq 0.80$ ).

For the hierarchical conditions (two or more second order factors), PRISM had comparable or better dimensionality accuracy than all other methods. The exception was with the largest sample size ( $N = 10,000$ ), where PApca had comparable or slightly higher accuracy in all conditions. PRISM achieved approximately 90% accuracy or better in every condition once  $N \geq 2,500$ . Lower order Louvain was the second best performing method up to very large samples ( $N \leq 5,000$ ), but was always just a step below PRISM. PApca was generally less accurate than the lower order Louvain and this difference increased as the number of second order factors and first order factors per second order factor increased. PApca was only comparable

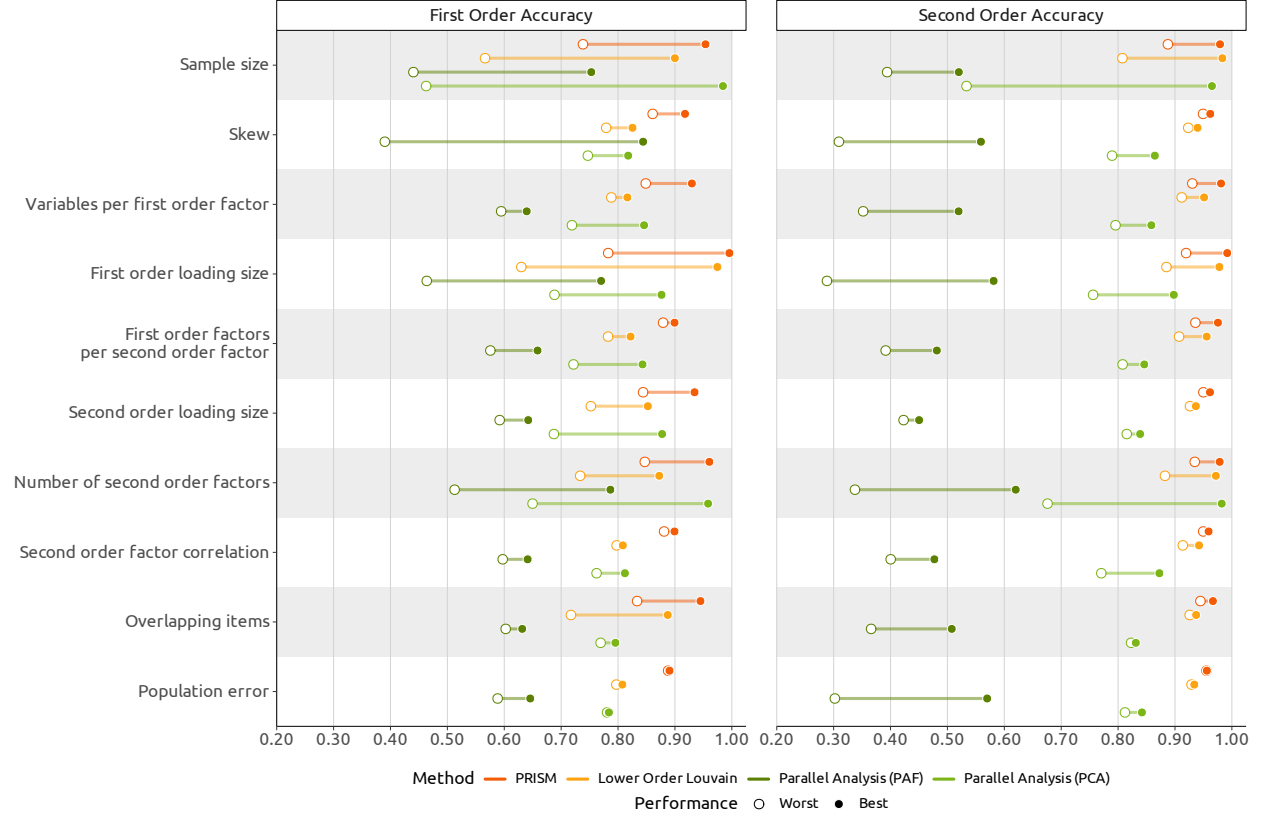
to lower order Louvain when there were two second order factors and three first order factors per second order factor. PApaf was consistently the worst performing method, rarely achieving accuracy greater than 80% even in exceptionally large sample sizes ( $N = 10,000$ ). Structural recovery improved with sample size for all methods with PRISM and lower order Louvain achieving at least good structural recovery regardless of sample size ( $\omega \geq 0.80$ ). PApca, however, did not achieve good structural recovery when there were five first order factors per second order factor and smaller sample sizes ( $N \leq 1,000$ ).

**Figure 4**  
*Second Order Accuracy and Structural Recovery*



*Note.* Second Order = number of second order factors, First Order = number of first order factors per second order factor, and  $\omega$  = Omega index.  $\omega$  is equivalent to accuracy when there is one second order factor.

Recovery for the second order factors generally followed the recovery of the first order factors (Figure 4). PRISM was comparable or better than all other methods, generally achieving at least 80% dimensionality accuracy regardless of sample size and structure. Lower order Louvain was the second best method but tended to perform considerably worse with smaller sizes ( $N \leq 1,000$ ) relative to PRISM. PApca performed as well as PRISM when there was one second order factor but its dimensionality accuracy decreased and grew to be much worse than PRISM and lower order Louvain as the number of second order factors increased. PApaf was, by far, the worse performing method, never achieving more than 75% accuracy with minimal improvement with larger sample sizes. PRISM and lower order Louvain had the best structural recovery, achieving good recovery ( $\geq 0.80$ ) regardless of sample size, and were generally comparable across conditions. In contrast, PApca and PApaf did not achieve good recovery until larger samples ( $\geq 2,500$ ) and their recovery decreased notably as the number of second order factors increased.

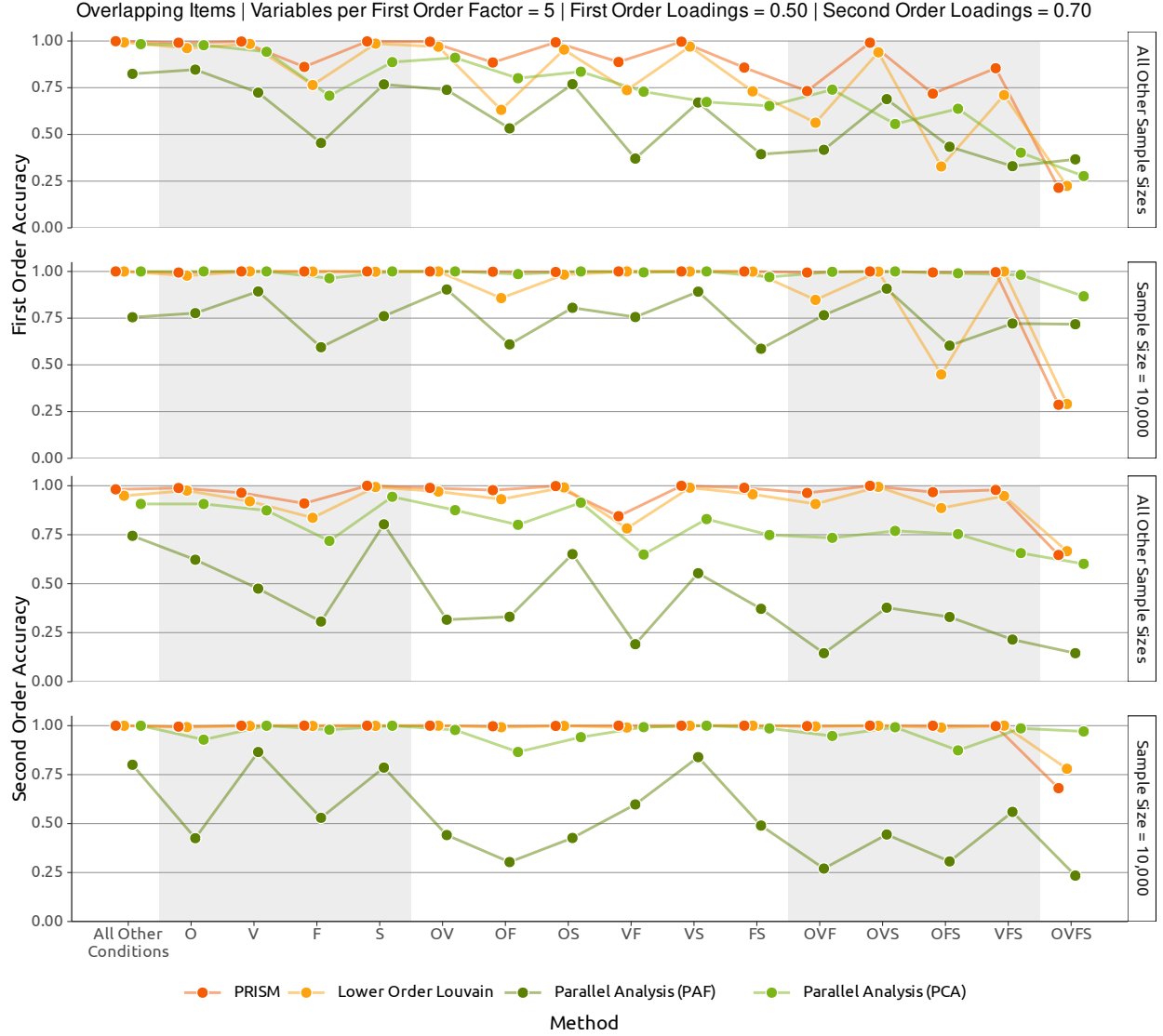
**Figure 5**
*Worst and Best Dimensionality Accuracy by Condition for First and Second Order Factors*


*Note.* y-axis starts at 0.20.

To understand the conditions each method was most affected by, its best and worst dimensionality accuracy for the first and second order factors were plotted by condition (Figure 5). Across all manipulated variables, PRISM was the least affected method, with differences ranging from 0.3% to 21.5% for the first order factors and from 0.1% to 9.2% for the second order factors. Its largest differences were due to sample size for the first and second order factors ( $\Delta 21.5\%$  and  $\Delta 9.2\%$ , respectively). Similarly, lower order Louvain and PApca were both most affected by sample size, for the first order ( $\Delta 33.4\%$  and  $\Delta 52.2\%$ , respectively) and second order factors ( $\Delta 17.8\%$  and  $\Delta 43.1\%$ , respectively). PApaf was most affected by skew for both the first order ( $\Delta 45.4\%$ ) and second order factors ( $\Delta 29.3\%$ ).

Across all conditions, there was a specific condition combination where all methods performed poorly (Figure 6). This condition combination was when overlapping dimensions were present, few variables per first order factor (5), low first order loadings (0.50), and high second order loadings (0.70). This condition combination was similarly difficult for all methods at  $N \leq 5,000$ . PApca was the exception where its dimensionality accuracy for the first and second order factors was high asymptotically ( $N = 10,000$ ). Outside of this specific condition combination, PRISM had comparable or better accuracy for the first and second order factors regardless of sample size.

**Figure 6**  
*Most Challenging Condition Combination*



*Note.* Results are split by first order and second order accuracy as well as by the largest sample size compared to all other sample sizes. O = overlapping dimensions, V = variables per first order factor (5), F = first order factor loadings (0.50), S = second order factor loadings (0.70)

Each of these conditions on their own and every pairwise combination were feasibly recovered by PRISM and Pafca when sample sizes were large enough ( $N = 10,000$ ), with accuracy near or above 90%. Lower order Louvain similarly recovered these conditions with the exception of overlapping dimensions combined with low first order loadings (accuracy around 85%). In smaller sample sizes, three-way combinations were more problematic, with PRISM dropping to around 75% first order accuracy when overlapping dimensions were combined with few variables per first order factor and low first order loadings (OVF) or with low first order loadings and high second order loadings (OFS). However, it was still the most accurate method in these conditions. Across all condition combinations, Pafca generally performed poorly for the first order and second order dimensionality accuracy regardless of sample size.

## 5 Discussion

The present study introduced the PRISM algorithm as a general purpose algorithm to identify hierarchical structures found in psychology. PRISM is a community detection procedure that identifies dimensions below the resolution limit of modularity (Fortunato & Barthélemy, 2007) by iteratively applying the standard Louvain algorithm to induced subgraphs. Using a Monte Carlo simulation study, PRISM was compared against several other exploratory methods for recovering second order factor structures under conditions expected to be common in psychology (e.g., ordinal variables, skew, overlapping dimensions). All methods were evaluated for first order factor recovery, with four (max loading, MAP, VSS1, and VSS2) chosen specifically to test their ability to set the depth of the popular bass-ackward procedure (Forbes, 2023; Girard, 2026; Goldberg, 2006). Generally, PRISM performed as well as or better than all other methods across conditions. hierEGA with lower order Louvain and PAPca performed well under certain circumstances such as the absence of overlapping dimensions and exceptionally large sample sizes, respectively. Recovery for second order factors was restricted to PRISM, lower order Louvain, PAPaf, and PAPca, with PRISM performing as well as or better than the other methods.

### 5.1 Recovering Structures with PRISM

The primary finding of our study is that PRISM generally outperformed all other methods in recovering both first and second order structures, including lower order Louvain, the previous best performing method. PRISM was also comparatively robust to skew, showing minimal declines in dimensionality accuracy and structural recovery from moderate to extreme skew relative to the other methods.

The relative performance of the comparison methods was consistent with the work by Jiménez et al. (2025) evaluating exploratory dimensionality estimation methods on hierarchical structures generated from bifactor models with multiple general factors. Consistent with their results, we found that lower order Louvain was generally comparable or more accurate than PAPca and substantially more accurate than PAPaf when recovering both lower (group or first order) and higher order (general or second order) structures. Our study extends this work by generating second order factor models with ordinal data, skew ranging from moderate to extreme, overlapping dimensions, and evaluation of structural recovery (Omega index). The consistency of our findings with their study suggests that the comparison methods provide a credible benchmark against which PRISM’s gains can be interpreted.

A particularly relevant finding was that PRISM achieved high dimensionality accuracy and structural recovery at much smaller sample sizes than other methods. Additionally, when samples were large ( $N = 2,500$ ), PRISM performed as well as or better than most methods did at much larger sample sizes, especially when there were many first order factors. This pattern held for the recovery of second order factors demonstrating that PRISM can recover structures at any level of the hierarchy (Samo et al., 2026). These findings suggest that PRISM is particularly well-suited for hierarchical studies with smaller samples that are typical in psychology (Huth et al., 2025).

One specific condition combination, however, presented difficulties for all methods including PRISM: overlapping dimensions combined with few items per first order factor, low first order loadings, and high second order loadings. Across the 18 conditions with this combination (5.6% of the 320 simulated conditions), first order dimensionality accuracy was around 40% or less for all methods, and second order accuracy was around 70% or less. Consistent with this pattern, all methods except PAPaf underfactored considerably more than in other conditions, suggesting that when first order factors share indicators, have weak primary loadings, and are strongly related through large second order loadings, the second order structure overshadows the narrower first order dimensions. This combination is expected to be relatively rare in practice because hierarchical constructs in psychology are generally composed of narrower, more strongly interrelated constructs (e.g., personality facets, psychopathological disorders) that are aggregated into broader, more heterogeneous domains (e.g., personality traits, psychopathological spectra; Forbes et al., 2024; Greene et al., 2023; Kotov et al., 2017; Schwaba et al., 2020; Widiger, 2017).

A notable condition for PRISM in our simulation was with a single second order factor, which is equivalent to a correlated factors model. The correlated factors model has been the structure most often used to evaluate community detection methods in network psychometrics (Christensen et al., 2024; Cosemans et al., 2021; Golino et al., 2020). Under these conditions, PRISM was comparable or more accurate than PAPca, the strongest traditional method in our simulation, with the clearest advantage in smaller samples and more first order factors. PRISM also rarely extracted more dimensions than were present, whereas lower order Louvain tended to overfactor. Taken together, these results are consistent with our expectation that PRISM

only recovers substructure when there is evidence of substructure, making it a robust approach to recover hierarchical and non-hierarchical structures.

## 5.2 Estimating the Depth of Bass-Ackward Hierarchies

A secondary aim of this simulation was to evaluate methods used to determine the maximum depth of bass-ackward hierarchies (Girard, 2026). The majority of the methods tested in this study consistently underfactored across conditions (Table 1), with underfactoring especially pronounced under extreme skew and overlapping dimensions. Both conditions are expected to be common in psychopathology data (Hoffman et al., 2019), where the bass-ackward method has often been applied (Faulkenberry et al., 2026; Forbes et al., 2025). The choice of dimensionality estimation method is therefore consequential because the bass-ackward method cannot recover levels beyond the maximum depth initially specified (Forbes, 2023).

The max loading criterion (Goldberg, 2006) was the only method evaluated that determined depth from the bass-ackward hierarchy itself (Tong et al., 2025). Although the max loading criterion generally outperformed MAP, VSS1, and VSS2 under moderate skew, its recovery of the first order structure deteriorated under extreme skew and as the number of first order factors increased. Based on our results, these traditional methods of dimensionality estimation are not suitable to set the depth of the bass-ackward method.

PAPca and PAPaf generally had better accuracy and recovery than the other traditional methods. PAPca exhibited modest bias with accuracy improving as sample size increased at both first and second order levels, making it the most viable bass-ackward depth-setting heuristic as well as hierarchical estimation method of the traditional dimensionality estimation methods we tested. At  $N = 10,000$ , PAPca was often comparable to PRISM and performed particularly well under the combination of conditions that all methods faltered (Figure 6). Notably, its accuracy was subpar in small-to-large sample sizes and with more first order factors, limiting its utility to exceptionally large sample sizes.

Despite relatively high dimensionality accuracy and structural recovery when skew was moderate, PAPaf consistently overestimated the number of first order factors, especially with extreme skew and population error (Table 1). In principle, some overfactoring may be preferable to underfactoring because the refinement steps of the extended bass-ackward method can remove redundant or artifactual dimensions (Forbes, 2023). The redundancy step, however, only removes dimensions that perpetuate across levels and retains its lowest level version, so overextracted factors at the bottom of the hierarchy are not eliminated. Removing these extra dimensions therefore requires the optional step of identifying artifactual dimensions, which is left to the researcher’s discretion. In practice, overfactoring should not be considered a safe error because removing the extra dimensions PAPaf produces depends on subjective judgment, which introduces researcher degrees of freedom and the risk of confirmation bias (Forbes, 2023).

Taken together, these findings indicate that many traditional dimensionality methods risk setting the initial depth of a bass-ackward hierarchy too shallow. The refinements applied in the extended bass-ackward method cannot correct underfactoring and may not adequately correct overfactoring. In light of our results, caution should be used when interpreting the existing bass-ackward hierarchies (Faulkenberry et al., 2026; Forbes et al., 2025; Kim & Eaton, 2015; Michelini et al., 2019).

## 5.3 Implications for Applied Researchers

Our findings support PRISM as a robust method to recover structures from the bottom up. PRISM provided the best overall recovery of both first and second order factors and was at least as accurate as the other methods in nearly all conditions. Its strong performance at smaller sample sizes make it particularly useful when exceptionally large samples are difficult to obtain such as clinical populations.

Importantly, using PRISM does not require a commitment to network theory (Borsboom, 2017). The utility of network variable models has generated considerable debate (Borsboom et al., 2017; Forbes et al., 2017, 2021; Fried et al., 2021), much of which has concerned the interpretation and replicability of individual network edges. The present study addresses a different question: whether network methods can accurately recover dimensional structure. A substantial body of research has demonstrated that network methods recover structures generated by latent factor models (Christensen et al., 2024; Cosemans et al., 2021; Golino et al., 2020; Goretzko, 2025; Jiménez et al., 2025), while recent work has shown that they also recover known communities generated by stochastic block models (Christensen & Choi, 2026). Collectively, this evidence suggests that the usefulness of network methods for structure recovery is not confined to a single data generating mechanism. Network methods can therefore be used as a general method for structure recovery independent of theoretical claims about whether the constructs themselves are networks.

PRISM can be used as a standalone procedure for constructing hierarchies from the bottom up. After identifying the most specific dimensions, their relations can be represented at the next level and PRISM can be applied recursively to identify progressively broader organization (Samo et al., 2026). In this way, the number and organization of dimensions at each level emerge from the same procedure rather than being prespecified or estimated using different methods across levels.

#### 5.4 Limitations

Although this simulation advances the evaluation of exploratory methods for recovering hierarchical structures, several limitations warrant further study. First, all data were generated from second order factor models. These models complement prior evaluations based on bifactor structures but do not represent the full range of hierarchies that may occur in psychological data, particularly in broad taxonomies such as personality and psychopathology. Such hierarchies may contain more than two levels or unequal numbers of indicators per dimension (Forbes & Wright, 2023; Samo et al., 2026). Alternative data generating models are also plausible, including hierarchical stochastic block models for network structures (Peixoto, 2014). Future simulations should therefore examine whether PRISM’s advantages generalize to deeper hierarchies and alternative data generating structures.

Second, the results are specific to the methods and implementations examined. All procedures used polychoric correlations in this study whereas empirical applications have often relied on Spearman correlations (e.g., Forbes et al., 2025, 2026). Prior research suggests that Spearman correlations can improve network estimation under skewed continuous data (Isvoranu & Epskamp, 2023), but whether they improve the dimensional structure estimation of the methods evaluated in this study remains unclear.

Finally, the range of dimensions used for the max loading method extended from two dimensions below the lowest estimate to two dimensions above the highest estimate produced by the non-network dimensionality estimation methods. This constraint could have prevented the method from identifying the ground truth when it fell outside that range; however, we found that the range contained the true number of first order factors in 97.1% of replicates. Excluding replicates that lacked the ground truth within the search range had a negligible effect on accuracy and  $\omega$  ( $\Delta < 0.01$ ) and bias ( $\Delta < |0.05|$ ).

#### 5.5 Conclusion

The accurate recovery of hierarchical structures is fundamental to how broad constructs are measured and assessed in psychology. Across a diverse set of conditions, PRISM had the most accurate recovery of first and second order structures, including at sample sizes where traditional methods were often inaccurate and biased. Importantly, PRISM is a general procedure that accurately recovers hierarchical and non-hierarchical structures. PRISM therefore allows the depth and organization of a hierarchy to emerge from the bottom-up, providing researchers with a robust approach to recover the structures that shape psychological theory, measurement, and classification.

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## 7 Disclosure Statement

The authors report there are no competing interests to declare.

## 8 Data Availability Statement

All data, code, and materials can be found on [GitHub](#).

## 9 CRediT Roles

Alexander P. Christensen: Conceptualization, Data Curation, Formal Analysis, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing - Original Draft, Writing - Review & Editing.

Sydney M. Goldberg: Conceptualization, Methodology, Writing - Review & Editing.

Ashley L. Watts: Conceptualization, Methodology, Writing - Review & Editing.

Luis Eduardo Garrido: Conceptualization, Formal Analysis, Investigation, Methodology, Validation, Writing - Review & Editing.

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